

PHYS0525 Analog and Digital Electronics – Spring 2026

Assignment 6

~~Due Monday, Mar 2, 2026~~

Due Friday Mar. 6 2026

1) A *npn* transistor has a $100\ \Omega$ resistor connected from its emitter to ground. The collector is connected directly to a 10 V supply. The base voltage is set to 1.5 V . and $\beta = 200$.

Draw a circuit diagram for the transistor circuit described above.

What is the voltage at the emitter?

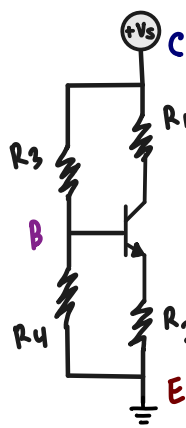
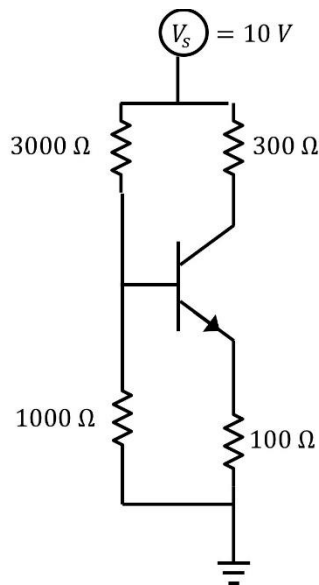
Is the transistor in active or saturation mode?

Determine the currents at the base, collector, and emitter.

$$V_{CE} = V_C - V_E = 10 - 0.9 = 9.1\text{ V}$$

↳ not near saturation ($V_{CE} > V_{BE}$): active

2) For the circuit drawn below, determine the voltages at the base, emitter, and collector of the transistor. Also determine the emitter, and collector currents. You may assume β is very large.



Assume I_B small

$$V_B = \frac{R_4}{R_3 + R_4} V_s$$

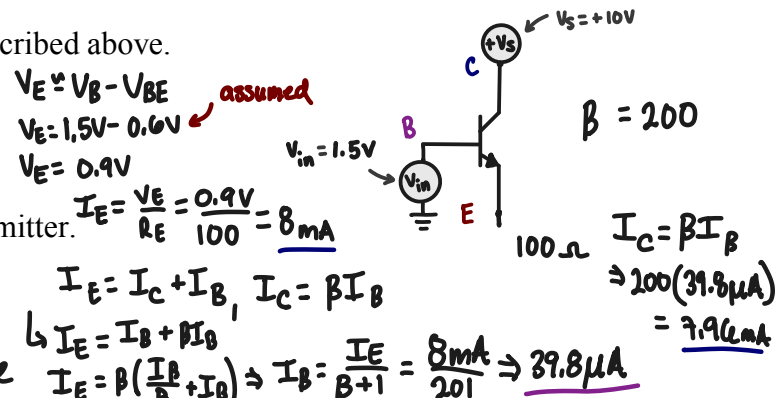
$$\Rightarrow V_B = \frac{1\text{ k}\Omega}{3\text{ k}\Omega + 1\text{ k}\Omega} V_s = \frac{1}{4} \cdot 10\text{ V} = 2.5\text{ V}$$

$$V_E = V_B - 0.6\text{ V} = 1.9\text{ V}$$

$$I_E = \frac{V_E}{R_2} \Rightarrow \frac{1.9\text{ V}}{100\ \Omega} = 19\text{ mA}$$

w/ β very large - $I_C \approx I_E = 19\text{ mA}$

$$* V_C = V_s - I_C (R_1) \Rightarrow 10\text{ V} - 19\text{ mA} (300\ \Omega) \Rightarrow 4.3\text{ V}$$



3) The two circuits shown below both act as followers. Circuit (a) is the simple emitter-follower that was discussed in class and in lab. Circuit (b) consists of two transistors connected in what is called a Darlington configuration (sometimes also referred to as a super-transistor). Together the transistors act as a follower, but with much larger input impedance than the single transistor emitter-follower. Assume R and V_s are chosen such that all transistors are in their active region of operation.

Convince yourself that the output of circuit (b) follows the input of circuit (b) by finding the voltages at the base of second transistor and the emitter of the second transistor. For each circuit there is an additional DC offset voltage at the output. We discussed in class that the offset voltage for circuit (a) is ≈ 0.6 V. What is the offset voltage for circuit (b)?

Assume that all the transistors have the same β value. What is the input impedance of circuit (a)? Assume that $V_{in} \gg 0.6$ V (Hint: Calculate the base current drawn from V_{in} . $Z_{in} = \frac{V_{in}}{I_{in}}$ or equivalently $Z_{in} = \frac{dV_{in}}{dI_{in}}$ for an input with an AC component.)

What is the input impedance of circuit (b)? Assume that $V_{in} \gg 1.2$ V and $R \gg \beta R_L$. (Hint: Calculate the base current drawn from V_{in} . $Z_{in} = \frac{V_{in}}{I_{in}}$ or equivalently $Z_{in} = \frac{dV_{in}}{dI_{in}}$ for an input with an AC component.) (Hint: The total emitter current of the first transistor is the sum of the current passing through R and the base current of the second transistor. If one of these is much smaller than the other you can ignore it).

